EPBA-Batch 14

Session 1- Data Analytics for Business

Anik Mukherjee

Management Information Systems Group

anik.mukherjee@iimcal.ac.in

LinkedIn Profile

IIMC Profile

INDIAN INSTITUTE OF MANAGEMENT 1961 CALCUTTA

Know Your Faculty



Anik Mukherjee

- **Assistant Professor, IIM Calcutta**
- **Ex-Faculty, IIT Roorkee**
- **MS-Ph.D., IIT Madras, Fulbright Fellow USA**
- **Research Areas:**
  - Computational Advertising
  - Cloud Computing
  - Healthcare Analytics
  - Cybersecurity
- **Teaching at IIM Cal:**
  - Cyber Security for Business
  - Blockchain for Business
  - Information Technology and Systems

Sessions (Sundays 9.00 AM to 12.00 AM)

Week	Session	Date	Online/On campus	Session	Module	Faculty
		16-Mar-2025	Online	Orientation	Orientation	Prof. Abhipsa & Prof. Anik
				Foundation Courses	Module:	
W1	1	23-Mar-2025	Online	Data Analytics for Business	**Foundation Courses**	Prof. Anik
W2	2	30 March 2025	Online	Introduction to Data Analytics using Case Pedagogy		Prof. Abhipsa
W3	3	6-Apr-2025	Online	Analytics and Digital Transformation		Prof. Abhipsa
W4	4	13-Apr-2025	Online	Introduction to Python for ML		Prof. Anik
W5	5	20-Apr-2025	Online	Introduction to Python: Applications in ML		Prof. Anik
W6	6	27-Apr-2025	Online	Statistics for Business Analytics - 1	Module:	Prof. Megha
W7	7	4-May-2025	Online	Statistics for Business Analytics - 2	**Statistics for Business Analytics**	Prof. Megha

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Faculties-Core-Modules-Course Wise



Prof Anik Mukherjee

- Asst Prof, IIM Calcutta
- Faculty, IIT Roorkee
- Ph.D., IIT Madras
- Areas: Analytics, Cloud Computing, Cybersecurity



Prof Abhipsa Pal

- Asst Prof, IIM Calcutta
- Faculty, IIM Kozhikode
- Ph.D., IIM Bangalore
- Developer, Infosys
- Areas: Fintech Adoption, Digital Transformation



Prof Megha Sharma

- Full Prof, IIM Calcutta
- Ph.D., IIM Ahmedabad
- Areas: Business data analytics, Statistics in analytics

Faculties-Core-Modules-Course Wise



Prof Soumyakanti Chakraborty

- Full Prof, IIM Calcutta
- Ph.D., IIM Calcutta
- Areas: Business data analytics, economics of IT, pricing of information goods



Prof Sumanta Basu

- Full Prof, IIM Calcutta
- Ph.D., IIM Ahmedabad
- Areas: Data analytics and Optimization; Pricing & Revenue Management



Prof Vimal Kumar M

- Assistant Prof, IIM Calcutta
- Ph.D., IIM Trichy
- Areas: Digital divide, data analytics

Faculties-Core-Modules-Application Area Wise



Prof Peeyush Mehta

- Full Prof, IIM Calcutta
- Ph.D., IIM Ahmedabad
- Areas: Supply chain, Bayesian games



Prof Randhir Kumar

Assistant Prof, IIM Calcutta

- Ph.D., University of Amsterdam
- Areas: Digital divide, data analytics



Prof Ritu Mehta

- Full Prof, IIM Calcutta
- Ph.D., IIT Kanpur
- Areas: Services marketing, consumer behavior



Prof Samit Paul

- Assistant Prof, IIM Calcutta
- Ph.D., IIM Lucknow
- Areas: Cost and managerial accounting; Financial Risk Management



Prof Anik Mukherjee

- Asst Prof, IIM Calcutta
- Faculty, IIT Roorkee
- Ph.D., IIT Madras
- Areas: Analytics, cloud computing, cybersecurity

Introduction to Analytics for managers

- AI/ML is a field of study that gives the computers the ability to learn from past historical data

- Example: Spam email detection, identifying fraudulent transactions, house price prediction, car arrival time prediction, sales prediction, weather forecasting and many others.

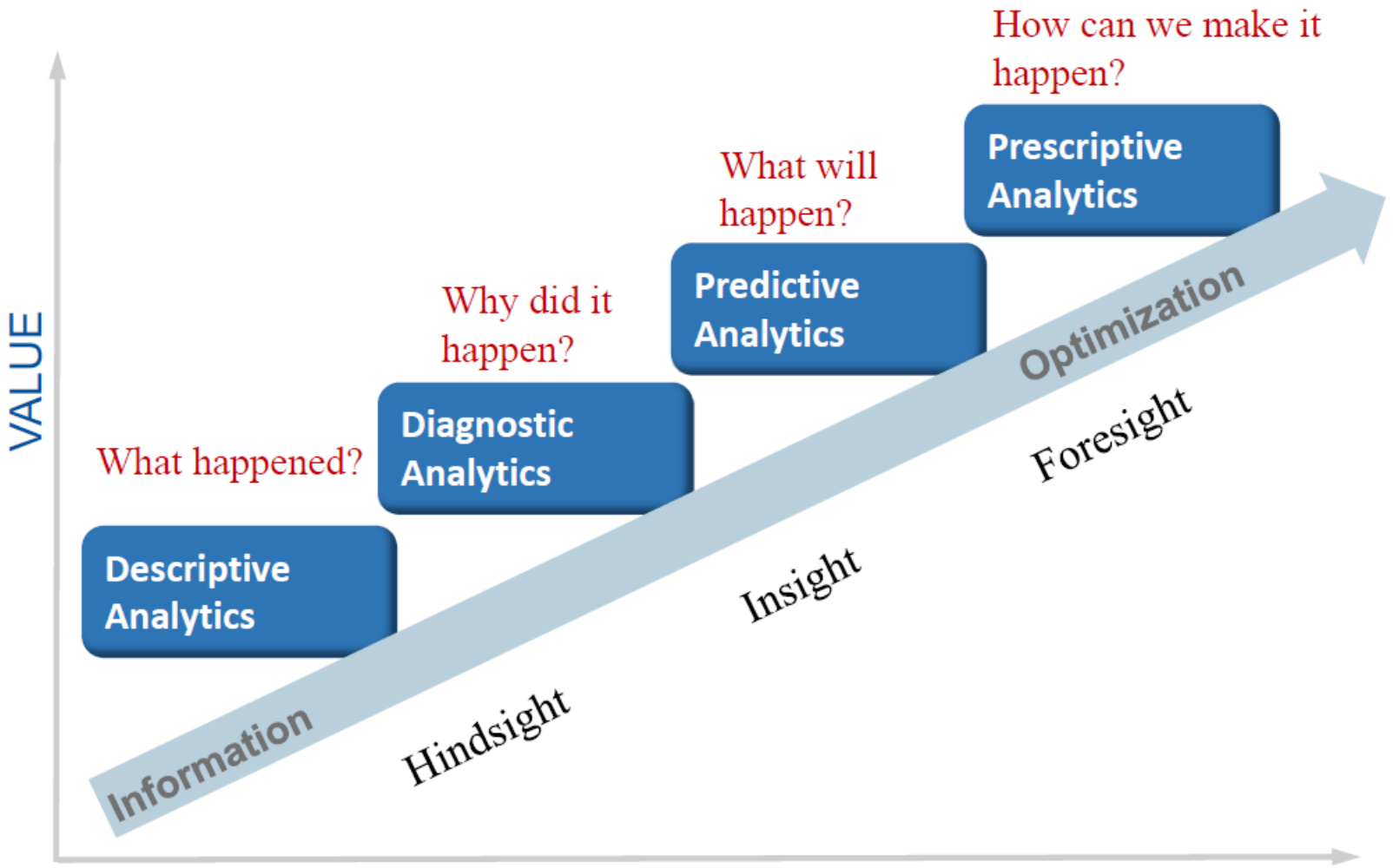
- AI/ML models are developed based on *Training Dataset* and evaluated on *Test Dataset* before deploying in production.

- Various performance measures of AI/ML models such as accuracy and others.

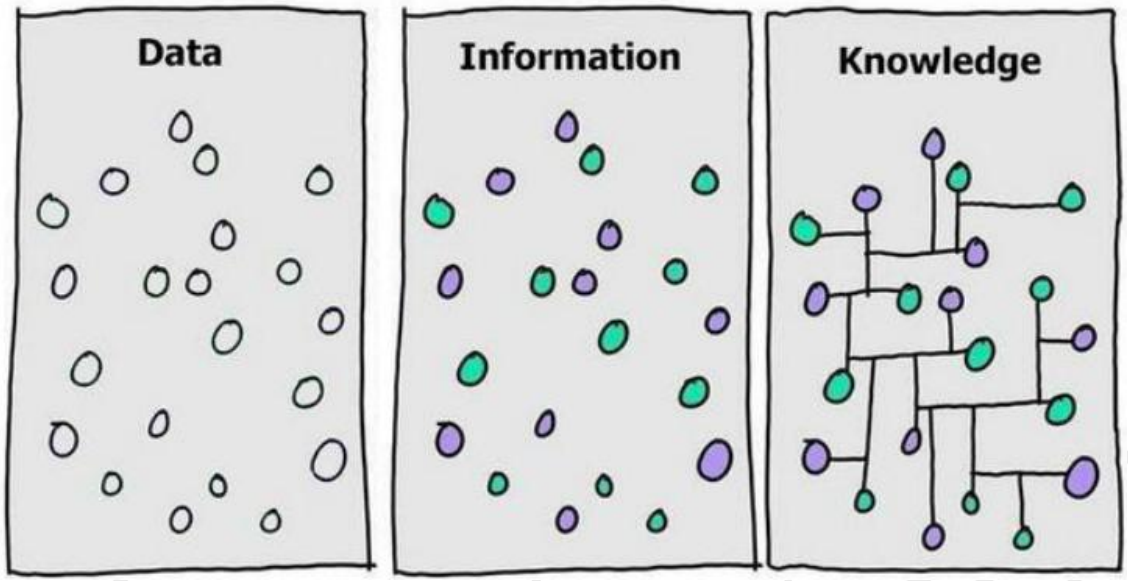
Why analytics in the filed of management?

- Data Science Managers/ Leads need to understand data science project life cycle
- Getting comfortable with data and understanding how to incorporate ML in your product
- Product analytics
- Understanding data related issues
- Data visualization

Analytical Model and Data Visualization



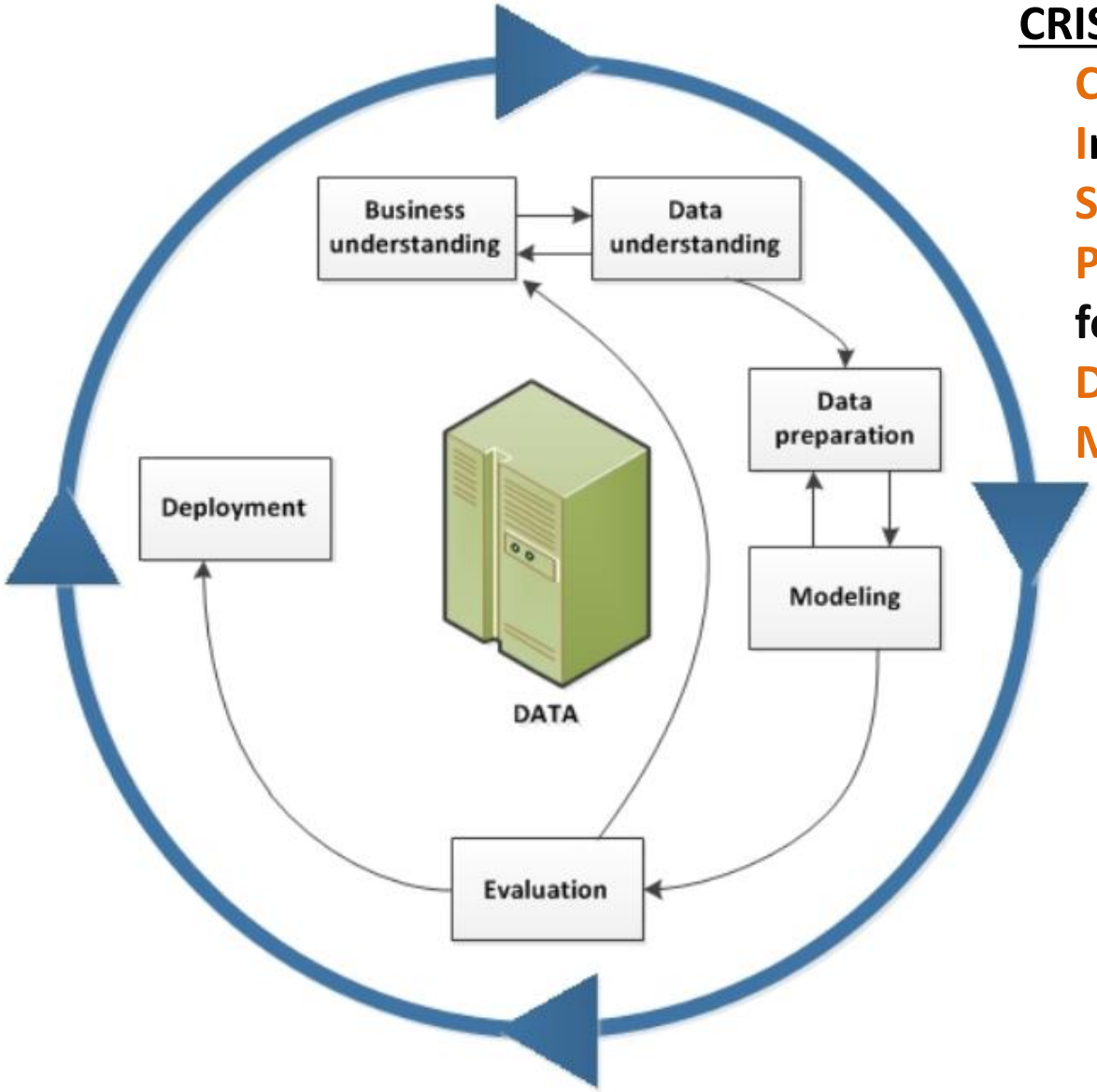
Data Vs Information vs Knowledge



5 V's of Big Data

- Volume (LLM- Large Language Model)
- Variety
  - Different types of data generated
- Velocity
  - Data being generated at an alarming rate
- Value
  - Mechanism to bring correct value out of it
- Veracity
  - Uncertainty and inconsistency in the data

Data Science Project Lifecycle



CRISP-DM Methodology

CRoss
Industry
Standard
Process
for
Data
Mining

Project Lifecycle cntd..

- Business Understanding
  - Determining business objective
  - Determining data mining goals
  - Prepare project plan
- Data Understanding
  - Collect data to meet business objectives
  - Check data quality i.e., missing data, data errors, coding inconsistencies, bad/ wrong filed name
- Data Preparation
  - Data formatting
  - Data cleaning/ removing noise
  - Integrating data from multiple sources to construct new data

Project Lifecycle cntd..

- Modelling
  - Selecting Modelling Techniques
  - Building the models
- Evaluation
  - Performance metrics
  - Compare competing models
- Deployment

EDA vs Data Visualization

- Exploratory Data Analysis

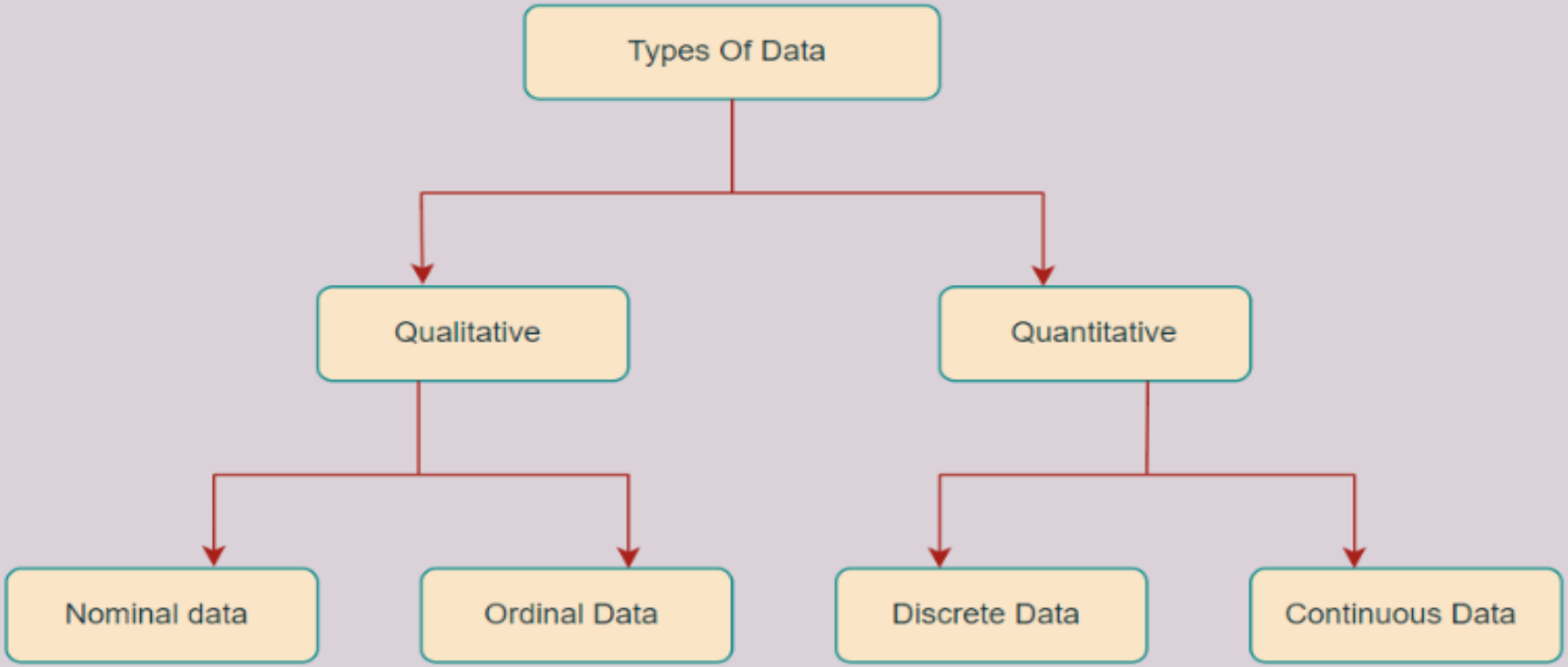
Missing Value Imputation

- EDA is the initial investigation of the data to find out patterns or anomalies

- Data visualization helps understand the underlying data patterns/ details

- Four R's of EDA:
  - Revelation: Data Visualization
  - Residuals: Difference between the actual and predicted
  - Re-Expression: Transform data
  - Resistance: Undue influence of outliers

Data Types and Data Visualization



- Colour
- Marital status
- Nationality
- Gender
- Eye Color

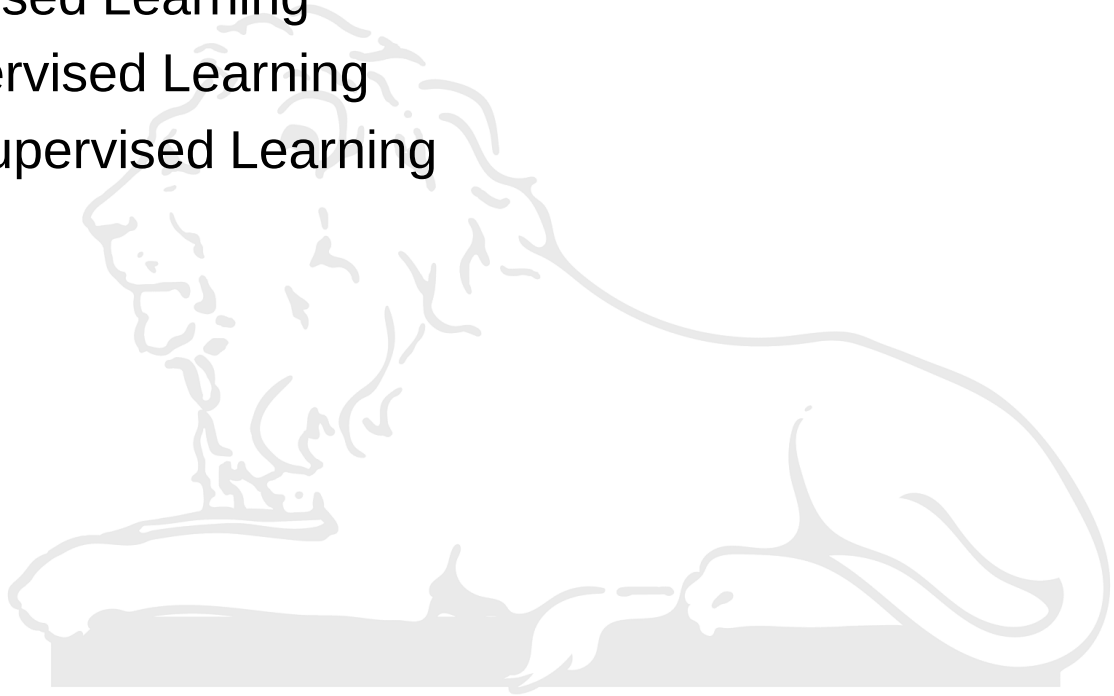
- When companies ask for feedback, experience, or satisfaction on a scale of 1 to 10
- Letter grades in the exam (A, B, C, D, etc.)
- Ranking of people in a competition (First, Second, Third, etc.)
- Economic Status (High, Medium, and Low)
- Education Level (Higher, Secondary, Primary)

- Total numbers of students present
- Cost of a cell phone
- Numbers of employees in a company
- The total number of players who participated in a competition
- Days in a week

- Height of a person
- Speed of a vehicle
- "Time-taken" to finish the work
- Wi-Fi Frequency
- Market share price

Types of Learning

- Can be classified into three main categories based on the amount of supervision required during *Training*:
  - Supervised Learning
  - Unsupervised Learning
  - Semi supervised Learning



Thank you ☺